CONFIDENTIAL

UTM UNIVERSITI TEKNOLOGI MALAYSIA

School of Computing,
Faculty of Engineering

RESEARCH UNIVERSITY

FINAL EXAMINATION
SEMESTER 2, 2019/2020

PART B2

SUBJECT CODE	:	**SCSR2043**
SUBJECT	:	**OPERATING SYSTEM**
YEARS / COURSE	:	**SCSI/J/R/V/D/B**
TIME	:	**70 MINUTES**
DATE	:	**15 JULY 2020**
PLACE	:	**ONLINE**

INSTRUCTIONS :

1. Answer **ALL** questions by writing on empty A4 paper;
2. For each page of your answer, write your **NAME**, **MATRIC NO.**, and **SECTION** of the subject (a template of answer sheet is provided separately);
3. Do snapshot or scan each page, convert into pdf, and compile as a single pdf file;
4. Upload the pdf file using the link provided in the e-learning.

This questions paper consists of **4** (**FOUR**) printed pages, excluding this page.

PART B2

STRUCTURED QUESTIONS (35 MARKS)

<u>QUESTION 3</u> [8 *Marks*]

Consider a portion of directory structure for a workstation as illustrated in Figure 3. Assume that a command prompt console is using to perform some operations using the UNIX command line and the current working directory is *outdoor* folder. Answer the following questions in sequence.

Figure 3

(*Source: https://stuyhsdesign.files.wordpress.com/2015/09/directory-structure1.png*)

(a) In the current directory, write a single command line to copy the *logo.png* into the *theater* folder. [3 *Marks*]

(b) In the current directory, write a single command line to change the working directory to *music* folder. [2 *Marks*]

(c) In the new current directory, write TWO different ways of single command line to set the access control of its files so that all classes can be read only. [3 *Marks*]

QUESTION 4 [16 Marks]

Consider the following Figure 4 illustrates a secondary disk space with a file stored in some blocks sequentially. Answer the following questions.

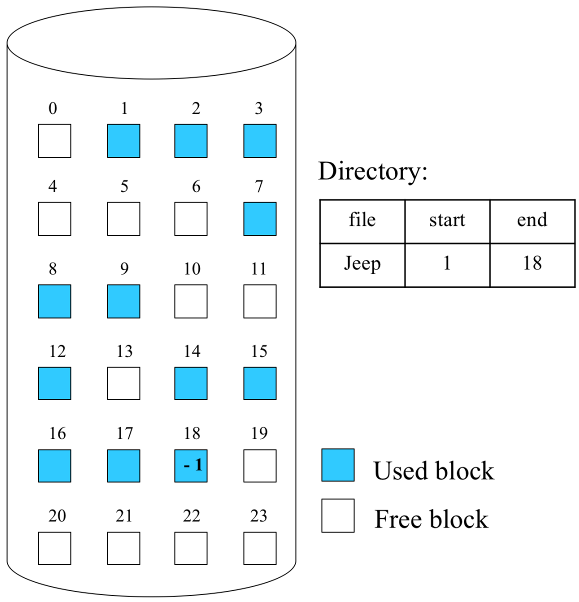

Figure 4

(a) Write the free space representation using the following methods clearly:

i) Grouping (assume each block can only store 3 block addresses) [2 *Marks*]

ii) Counting [2 *Marks*]

(b) If the capacity of storage is 98304 bytes, and the number of blocks as shown in Figure 4, what is the size of file *jeep* in bytes? Assume all blocks fully occupied. Show your working. [2 *Marks*].

(c) Consider the file *jeep* is edited, and requires some block of storage to be appended at the end of the file.

i) If the original size of the file *jeep* is 22.528 Kb that occupied all the blocks used as shown in Figure 4, how many additional blocks are required when the file size becomes 30.176 Kb? Show your working. [2 *Marks*]

ii) Based on the previous answer, state the block number that produced the internal fragmentation and calculate the size. Show your working. [2 *Marks*]

(d) While checking for block consistency, the result of scanning 16 blocks of storage is shown in Figure 5.

Block numbers	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Block in used	1	1	0	1	0	1	1	1	1	0	0	1	1	1	0	0
Free blocks	0	0	1	0	2	0	0	0	0	1	0	0	0	0	1	1

Figure 5

i) Based on the Figure 5, determine the consistency error(s) exist by stating to the block number of the storage. Justify your answer. [4 *Marks*]

ii) For the error(s) stated in previous answer, suggest a solution to overcome the error(s) [2 *Marks*]

<u>QUESTION 5</u> [11 *Marks*]

Consider the following information:

- ✓ A disk drive with 200 cylinders;
- ✓ A request queue :

  98, 149, 37, 122, 14, 124, 65, 67
- ✓ Head pointer 25.
- ✓ Direction: the disk arm is moving to innermost.

Answer the following questions.

(a) By using the C-LOOK disk scheduling algorithms, illustrates by drawing the movement of the disk arm to satisfy the requests. Label clearly. [7 *Marks*]

(b) Calculate the total distance (in cylinders) that the disk arm moves to satisfy the requests. Show your working. [2 *Marks*]

(c) If the LOOK disk scheduling algorithms is used, briefly describe a conclusion that can be made about the disk arm movement to satisfy the requests. [2 *Marks*]

------- End of Question -------

GOOD LUCK !